

Task A — Playwright: Web E2E (Hands-on) — FINAL

Khodro45 — Senior Automation Engineer Interview (Round 2)

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Scope & Target Sites

Candidate must implement Playwright automation focused on the following demo websites:

- Forms section on <https://demoqa.com/forms>
- Book Store Application on <https://demoqa.com/books> — specifically: Login flow + Book Store interactions (search/add/remove books).

Core Focus Areas (must be demonstrated)

1. Understanding of Automation (web, software engineering concepts, reporting).
2. Completing full setup documentation (README.md).
3. Dockerize project and implement CI/CD.
4. Git workflow (feature branches, PRs, atomic commits).

Deliverables (explicit)

- Playwright TypeScript test suite (or Python) covering: demoqa Forms scenarios (form submission validations, file upload, nested forms) and Book Store: login + add/remove books + search.
- Page Object Model structure and reusable helpers (auth, fixtures, test data).
- README.md with full setup, env variables, how to run locally and in CI, Docker instructions, and troubleshooting.
- Dockerfile and (optional) docker-compose.yml to run tests in CI and locally.
- GitHub Actions workflow YAML that builds image, runs tests (matrix: chromium/firefox), uploads artifacts (HTML report, screenshots), and publishes a JUnit/HTML report.
- Branch: work must be submitted in a feature branch; include PR with clear description and screenshots of test run.
- Short design note explaining selector strategy, anti-flakiness measures, parallelization approach, and reporting strategy.

Evaluation Rubric (detailed)

Technical (50%): - Correctness & coverage of web flows — 20% - Robustness & anti-flakiness — 15% - Docker & CI integration (runs in container, artifacts) — 10% - Reporting & evidence (HTML/JUnit reports, screenshots) — 5%

Engineering Practices (30%): - README completeness & reproducibility — 10% - Git workflow & PR hygiene — 10% - Code structure, modularity & reuse — 10%

Soft / Ownership (20%): - Documentation quality & troubleshooting guidance — 10% - Communication in PR and change notes — 10%

Minimum Passing Requirements

- Tests running and passing in CI for at least Chromium headless.
- Dockerized runner that can execute tests locally using provided README instructions.
- README complete and clear (run steps + env + CI).
- PR demonstrating git workflow and atomic commits.

Suggested Project Layout

```
/project-root
  /tests
  /pages
  /fixtures
  /reports
  package.json
```

```
playwright.config.ts
Dockerfile
README.md
.github/workflows/ci.yml
```

Sample Dockerfile notes (candidate must provide actual file)

```
FROM node:18-bullseye-slim
RUN apt-get update && apt-get install -y wget ca-certificates && rm -rf /var/lib/apt/lists/*
WORKDIR /usr/src/app
COPY package*.json ./
RUN npm ci
COPY . .
RUN npx playwright install --with-deps
CMD [ "npm", "run", "test:ci" ]
```

Git Workflow Expectations

- Branch from main: feature/automation/demoqa-forms or feature/automation/demoqa-bookstore - Make small, logical commits with messages like: 'tests(forms): add validation scenarios for practice form' - Open PR to main with description, test evidence, and runbook - Use labels and reviewers; squash merge permitted after approval

Reviewer checklist

- Repo with tests and README.md
- Dockerfile & instructions
- CI workflow that runs tests and uploads artifacts
- Playwright tests covering Forms + Book Store
- PR with feature branch and commit history
- Reports (HTML/JUnit) and screenshots included